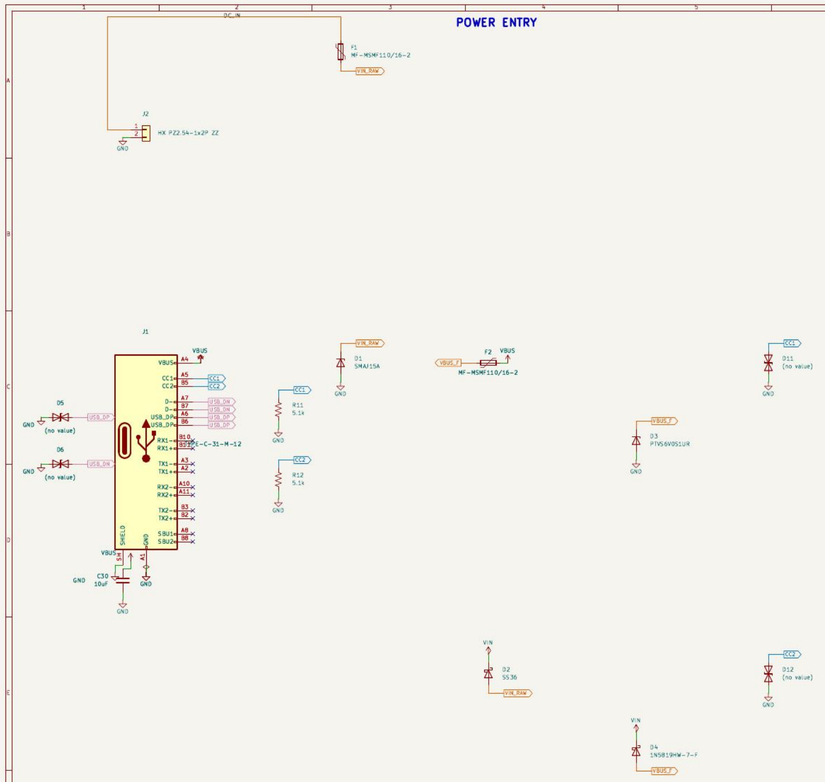
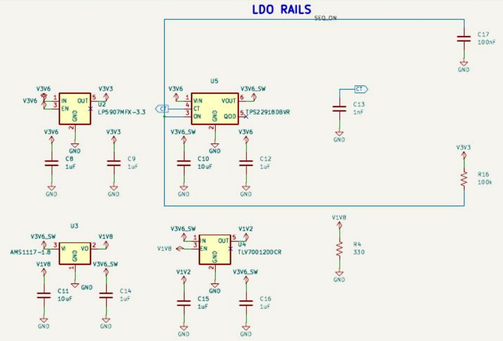
POWER ENTRY



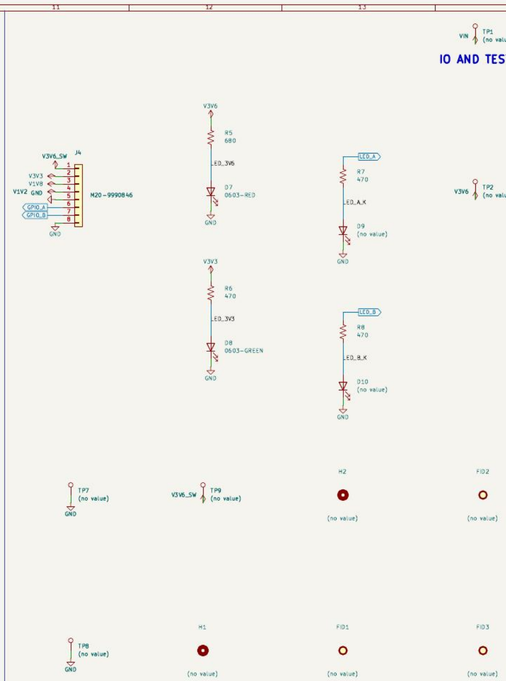
DC input 5–12 V. PTC F1. TVS D1 (15 V standoff, 24.4 V clamp). SS36 D2 reverse block.
USB VBUS: PTC F2. TVS D3. 1N5819 D4 ORs VBUS into VIN.

LDO RAILS



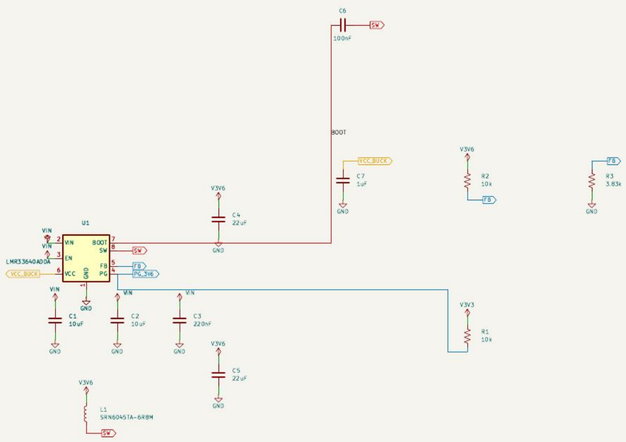
3V3 LP5907 from 3V6; 1V8 AMS1117 (5 V in its minimum load); 1V2 TLV70012.
Sequence: 3V3 up, RSS/C17 delay ~10 ms, TPS22918 passes 3V6 to the 1V8 LDO, 1V8 enables the 1V2 LDO.

IO AND TEST



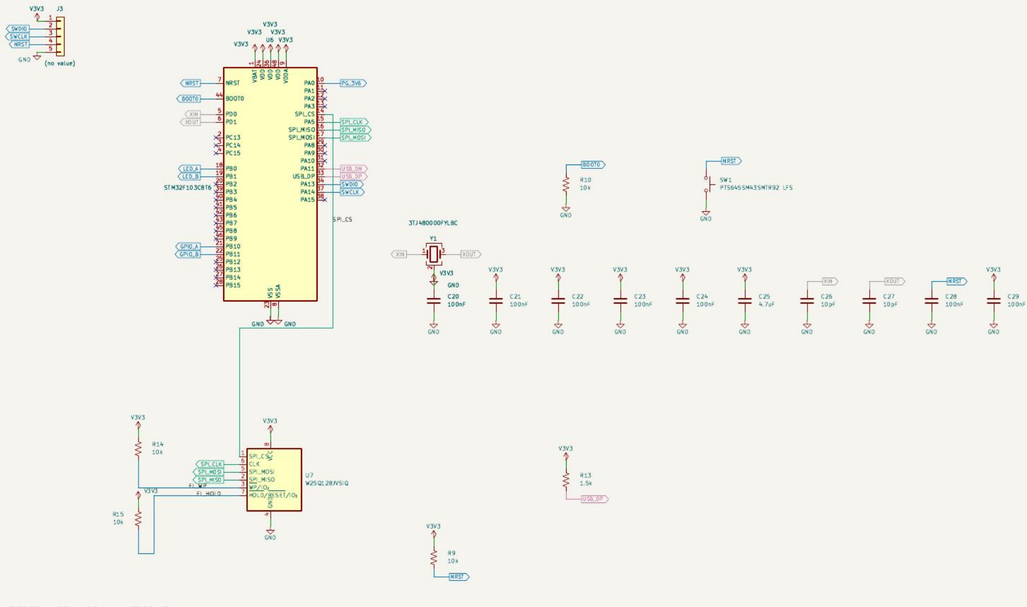
J4 carries 3V6_SW, 3V3, 1V8, 1V2 and two GPIO. TP1 VIN, TP2 3V6, TP3 3V3, TP4 1V8, TP5 1V2, TP6 VBUS, TP7–8 GND, TP9 3V6_SW.

BUCK 3V6



LMR33640 VFB 1.0 V: Vout = 1.0 × (1 + 10k / 3.83k) ≈ 3.611 V.
Internal compensation: no external network. L1 6.8 uH, ripple 0.93 A at 12 V in.

COMPUTE CORE



STM32F103CB needs 3.3 V only; 8 MHz crystal for USB; SWD on J3.

Fab: jlc-filled-via
86 components / 42 nets
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File: multirail_compute_node.kicad_sch

Title: multirail_compute_node

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KiCad E.D.A. 10.0.6		id: 1/1